

R4 Newton's third law

Name: _____

Class: _____

Date: _____

Time: **210 minutes**

Marks: **173 marks**

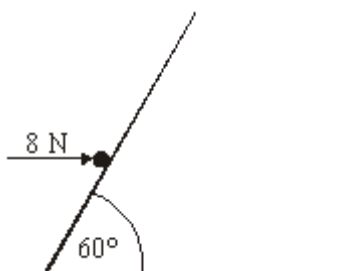
Comments:

EXAM PAPERS PRACTICE

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Q1.

A particle is held at rest on a smooth slope by a horizontal force of magnitude 8 newtons, as shown in the diagram below. The slope is at an angle of 60° to the horizontal.



- (a) Draw a diagram to show the forces acting on the particle. (1)
 - (b) By resolving horizontally show that the magnitude of the normal reaction force acting on the particle is approximately 9.24 newtons. (3)
 - (c) Find the mass of the particle, giving your answer to two significant figures. (3)
- (Total 7 marks)

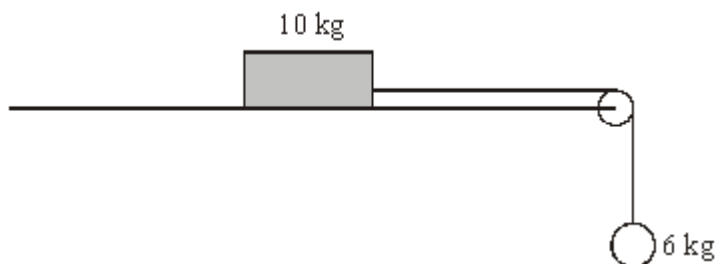
Q2.

A box, of mass 20 kg, is initially at rest on a rough horizontal surface. A horizontal force of magnitude P newtons is applied to the box. The coefficient of friction between the box and the surface is 0.3.

- (a) State the magnitude of the normal reaction force acting on the box. (1)
 - (b) Find the magnitude of the friction force that acts on the box if:
 - (i) $P = 80$; (3)
 - (ii) $P = 40$. (3)
 - (c) Find the value of P when the box is accelerating at 0.8 m s^{-2} . (3)
 - (d) When the box reaches a speed of 6 m s^{-1} , the horizontal force P is removed. Find the distance that the box travels after the force P is removed. (5)
- (Total 12 marks)

Q3.

A block, of mass 10 kg, rests on a rough horizontal surface. It is connected by a light, inextensible string to a particle of mass 6 kg. The string passes over a light, smooth pulley, so that the string hangs vertically, as shown in the diagram.



Model the block as a particle.

- (a) The system is released from rest and the block travels 0.5 metres in 2 seconds. Find the acceleration of the system.

(2)

- (b) Show that the tension in the string is 57.3 N.

(3)

- (c) Find the coefficient of friction between the block and the plane.

(6)

(Total 11 marks)

Q4.

A particle of mass 60 kg is on a rough surface inclined at an angle of 40° to the horizontal.

- (a) Find the magnitude of the normal reaction force acting on the particle.

(2)

- (b) If the particle remains at rest, find the minimum value of the coefficient of friction between the particle and the slope.

(4)

- (c) If the coefficient of friction between the particle and the slope is 0.2, the particle slides down the slope. Find the acceleration of the particle in this case.

(5)

(Total 11 marks)

Q5.

A block of wood has mass 4 kg. It is placed on a rough horizontal surface and is pulled by a horizontal string. The coefficient of friction between the block and the surface is 0.4.

- (a) Draw a diagram to show the forces acting on the block.

(1)

- (b) Calculate the magnitude of the normal reaction force acting on the block.

(1)

- (c) If the acceleration of the block is 2 m s^{-2} , find the tension in the string.

(3)

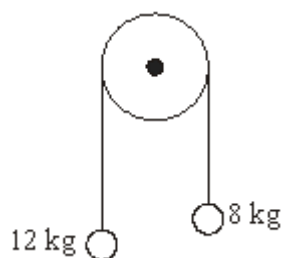
- (d) If the tension in the string is 20 N , find the acceleration of the block.

(2)

(Total 7 marks)

Q6.

A light, inextensible string has a particle of mass 8 kg attached to one end and a particle of mass 12 kg attached to the other end. The string passes over a smooth, light pulley. The particles are released from rest with the string taut and vertical on each side of the pulley. The diagram shows the pulley and the particles.



- (a) By forming an equation of motion for each particle, show that the acceleration of the particles is 1.96 m s^{-2} .

(5)

- (b) Find the tension in the string.

(2)

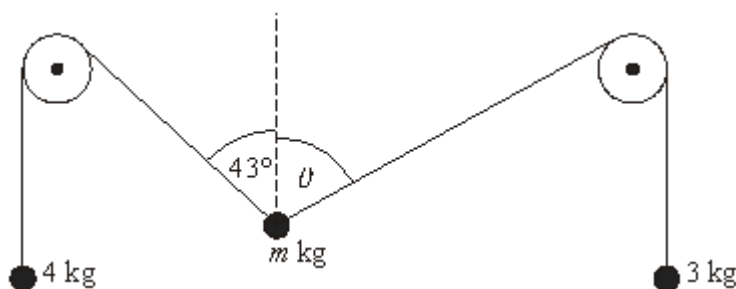
- (c) Find the time that it takes for the particles to reach a speed of 7 m s^{-1} .

(2)

(Total 9 marks)

Q7.

Two light, inextensible strings are attached to a particle of mass $m \text{ kg}$. Each string passes over a fixed, smooth, light pulley. The other end of one string is attached to a particle of mass 4 kg . The other end of the second string is attached to a particle of mass 3 kg . The diagram shows the system in its equilibrium position. The angles marked on the diagram are between the strings and the vertical.



- (a) Calculate the tension in each of the strings.

(2)

- (b) Show that $\theta = 65.4^\circ$, correct to three significant figures.

(5)

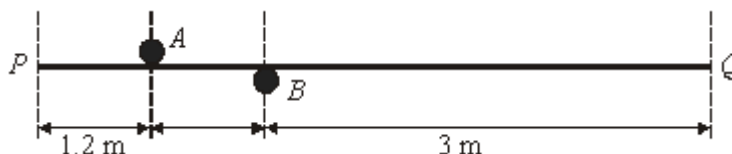
- (c) Find m .

(4)

(Total 11 marks)

Q8.

A uniform beam, PQ , has length 5 metres and mass 40 kg. It is placed between two fixed horizontal bars, A and B , so that the beam remains horizontal, as shown in the diagram.



- (a) Draw a diagram to show the forces acting on the beam.

(1)

- (b) Show that the force exerted on the beam by bar A has magnitude 245 N.

(3)

- (c) Find the magnitude of the force exerted on the beam by bar B .

(2)

- (d) An object of mass 5 kg is placed on the beam at Q . Find the forces now exerted on the beam by the bars A and B .

(6)

(Total 12 marks)

Q9.

Two forces $\mathbf{F}_1 = (3\mathbf{i} + 4\mathbf{j})\text{N}$ and $\mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})\text{N}$, act on a particle. The resultant of these two forces is $\mathbf{F}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Find $\mathbf{F}$.

(2)

- (b) Find the magnitude of $\mathbf{F}$.

(2)

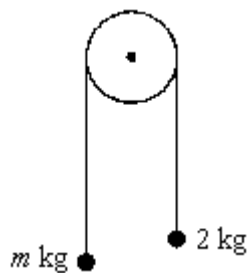
- (c) Find the acute angle between $\mathbf{F}$ and the unit vector $\mathbf{i}$.

(3)

(Total 7 marks)

Q10.

Two particles of mass 2 kg and mass m kg are connected by a light inextensible string that passes over a smooth light pulley, as shown in the diagram below.



The particles are released from rest with the string taut.

- (a) The 2 kg particle moves vertically downwards, travelling 0.8 metres in 2 seconds. Find its acceleration.

(3)

- (b) Find the tension in the string.

(3)

- (c) Find m , giving your answer to two significant figures.

(4)

(Total 10 marks)

Q11.

A block, of mass 5 kg, is held at rest on a rough plane, which is inclined at 30° to the horizontal. The block is released and slides down the plane. The coefficient of friction between the block and the plane is 0.2.

- (a) Draw a diagram to show the forces acting on the block as it slides.

(1)

- (b) Show that the magnitude of the friction force acting on the block is approximately 8.5 N.

(3)

- (c) Find the acceleration of the block.

(4)

- (d) Find the speed of the block when it has travelled 1.2 metres down the slope.

(3)

(Total 11 marks)

Q12.

A skier slides in a straight line directly down a slope inclined at 30° to the horizontal. The coefficient of friction between her skis and the slope is 0.3. The skier and her equipment are to be modelled as a particle of mass 80 kg. Assume that there is no air resistance present.

- (a) Draw a diagram to show the forces acting on the skier.

(1)

- (b) (i) Find the magnitude of the normal reaction force acting on the skier.

(2)

- (ii) Show that the magnitude of the friction force acting on the skier is 204 N to three significant figures.

(2)

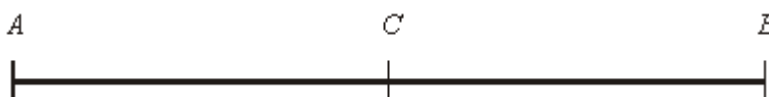
- (c) Find the acceleration of the skier.

(4)

(Total 9 marks)

Q13.

A uniform rod, AB , has length 4 metres, and C is the mid-point of AB .



Three particles are attached to the rod:

- one of mass 7 kg at A ;
- one of mass 12 kg at C ;
- and one of mass 11 kg at B .

The mass of the uniform rod AB is 10 kg.

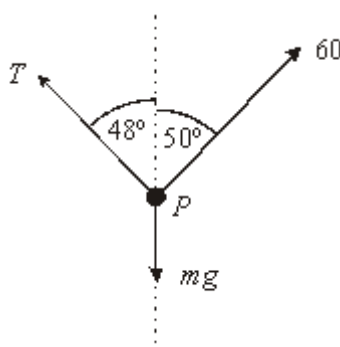
The rod rests in a horizontal position supported at A and B .

Find the magnitudes of the vertical forces exerted by the supports on the rod at A and B .

(Total 5 marks)

Q14.

A particle, P , of mass m kg, is held in equilibrium by two strings. One string is inclined at 50° to the vertical and exerts a force of 60 newtons on the particle. The other string exerts a force of magnitude T newtons at an angle of 48° to the vertical. The forces that act on the particle are shown in the diagram below.



- (a) Find T .

(3)

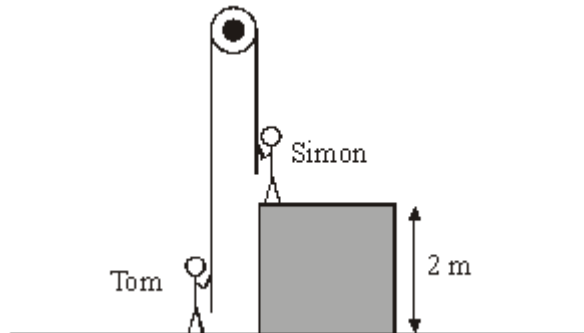
(b) Find m .

(4)

(Total 7 marks)

Q15.

Two children are holding the ends of a light, inextensible rope, which passes over a light, smooth pulley. Initially Tom, who has a mass of 40 kg, is standing at ground level and Simon, who has a mass of 60 kg, is on the edge of a fixed platform 2 metres above ground level. Model the two boys as particles, one initially at ground level, and the other initially at a height of 2 metres. The rope is taut.



Simon steps off the platform and as he falls vertically, Tom rises vertically.

(a) Assume that the rope remains taut while the boys are moving.

(i) Show that the acceleration of each boy is 1.96 m s^{-2} .

(5)

(ii) Find the tension in the rope.

(2)

(b) Find the total distance that Tom travels upwards.

(7)

(Total 14 marks)

Q16.

A particle moves on a horizontal plane subject to the following three forces:

$$\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j})\text{N}, \quad \mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})\text{N} \quad \text{and} \quad \mathbf{F}_3 = (-2\mathbf{i} + 10\mathbf{j})\text{N}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

Do not include any other forces in your calculations.

(a) Find the resultant force on the particle.

(2)

(b) The mass of the particle is 8 kg. Find the acceleration of the particle.

(1)

(c) At time $t = 0$, the particle is at the origin and has velocity $(3\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$.

(i) Find the velocity of the particle when $t = 40$ seconds.

(2)

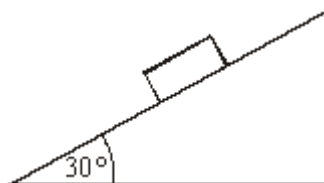
(ii) Find the distance of the particle from the origin when $t = 40$ seconds.

(5)

(Total 10 marks)

Q17.

A block, of mass 7 kg, is placed on a rough slope that is inclined at 30° to the horizontal, as shown in the diagram. The block remains at rest in this position.



(a) Draw a diagram to show the forces acting on the block.

(1)

(b) Find the magnitude of the normal reaction force acting on the block.

(2)

(c) Find the magnitude of the friction force acting on the block.

(2)

(d) The coefficient of friction between the block and the plane is μ . Find an inequality that μ must satisfy.

(2)

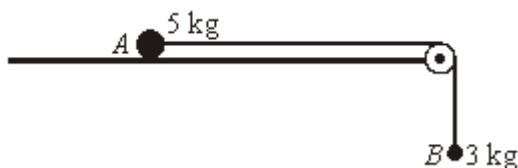
(e) A similar block, of mass 14 kg, is placed on the slope. Does this block remain at rest or slide? Give a reason for your answer.

(2)

(Total 9 marks)

Q18.

The diagram shows two particles that are connected by a light inextensible string. The 5 kg particle, A , is on a rough horizontal surface and the 3 kg particle, B , hangs on the other end of the string as shown in the diagram. The string passes over a smooth light pulley.



The coefficient of friction between A and the horizontal surface is 0.8. The particles are set in motion with the string taut and A moving towards the pulley. The initial speed of both particles is 2 m s^{-1} . The particles slow down and stop before A reaches the pulley or B hits the floor.

- (a) Show that, while both particles are moving, the magnitude of their acceleration is 1.225 m s^{-2} .

(6)

- (b) Find the tension in the string while the particles are moving.

(2)

- (c) Find the distance that the particles move before they come to rest.

(3)

(Total 11 marks)



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